Introduction to Special Relativity

October 30, 2024

Most of the information included in this handout is not covered thoroughly in lecture or in the homework assignments and therefore not important for you to memorize or practice for the sake of the exams in this course. However, this document is meant to give you a chance to, if curious, take a closer look at the derivations and concepts discussed in recitation this week (week 6).

The main idea of special relativity is simply that the laws of physics are **identical** in all inertial reference frames. An **inertial reference frame** is simply a frame of reference that is moving with a constant velocity. You can feel this when moving at a constant speed in a car or train – drop a ball and it will accelerate downwards, exactly as if you were standing still on the curb. The difference between this realization and special relativity however, is that the speed of light must also be the same, even if you're moving close to the speed of light.

Part 1 – Derivation of Time-Dilation from First Principles:

In the previous paragraph, I mentioned a simple idea: the speed of light is the same in all reference frames. This alone is enough to derive the major results of special relativity. Here, we will use this fact to derive the concept of time-dilation.

Imagine sitting on a ship or train that is moving very fast. You shoot a laser off of the ceiling (height **h** off the ground) and measure how long it takes to come back. Of course, it is moving at the speed of light, **c**.

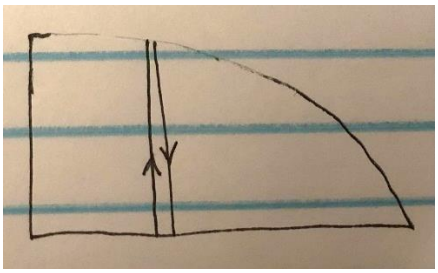


Figure 1: Light travels twice the height of the ship in the moving frame.

The diagram above is a crude representation of this. The total distance traveled by the light is:

$$\boldsymbol{d = 2h}$$

And the change in time for the beam of light to return to the floor is (Note that the prime in the $\boldsymbol{\Delta t'}$ simply indicates that we're referring to the moving frame – inside the train) :

$$\Delta t' = \frac{d}{c} = \frac{2h}{c}$$

Now let's move on to the moving frame. Since the speed of light is the same, we can immediately tell that the light should take longer to reach the floor when viewing as a static, external observer:

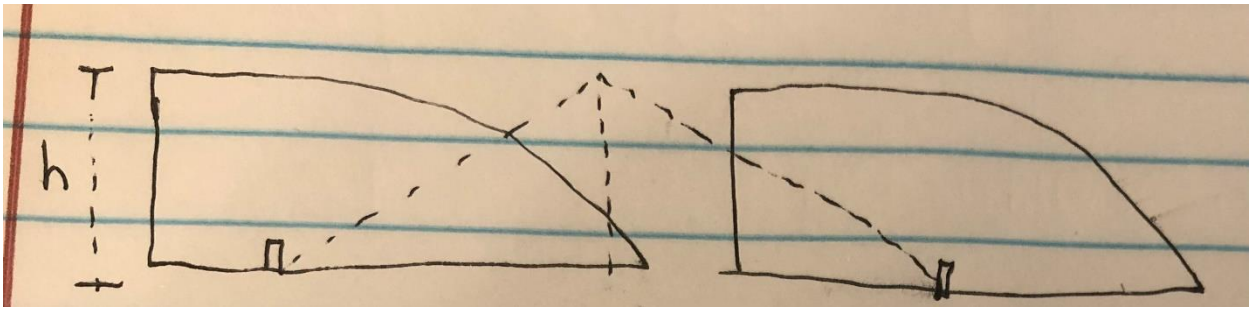


Figure 2: According to the outside observer, light travels a greater distance. How is that possible if the speed of light is the same? The Answer: time dilation.

The light beam travels the vertical distance, $2h$, as well as the horizontal distance, $v\Delta t$ where v is the speed of the train car and Δt is the change in time from the perspective of the outside observer. The distance traveled will therefore be:

$$d = \sqrt{(2h)^2 + (v\Delta t)^2}$$

Again, our Δt will be equal to the distance travelled divided by the speed of light:

$$\Delta t = \frac{d}{c} = \frac{\sqrt{(2h)^2 + (v\Delta t)^2}}{c}$$

This isn't useful yet, because both sides of the equation have the variable Δt. To fix this, we will have to do some algebra. What follows is the solution for Δt step by step:

$$\Delta t = \frac{\sqrt{(2h)^2 + (v\Delta t)^2}}{c}$$

$$c\Delta t = \sqrt{(2h)^2 + (v\Delta t)^2}$$

$$c^2\Delta t^2 = 4h^2 + v^2\Delta t^2$$

$$\Delta t^2(c^2 - v^2) = 4h^2$$

$$\Delta t^2 = \frac{4h^2}{c^2 - v^2}$$

$$\Delta t^2 = \frac{4h^2/c^2}{1 - (\frac{v}{c})^2}$$

$$\Delta t = \frac{\frac{2h}{c}}{\sqrt{1 - \frac{v^2}{c^2}}}$$

Take a deep breath. That's the end of our derivation. Now we just have some simplifying to do. You may remember:

$$\gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}$$

AND (from earlier):

$$\Delta t' = \frac{2h}{c}$$

Which means (if we substitute these into the second equation on this page):

$$\Delta t = \gamma \Delta t'$$

OR ALTERNATIVELY:

$$\Delta t' = \frac{\Delta t}{\gamma}$$

Which is the form of time dilation we're using in class today. Time moves at a slower rate for those in the moving reference frame when compared to outside observers.

Part 2 – Finding velocity from relativistic momentum:

You might remember that the common equation $\vec{F} = m\vec{a}$ is only approximately true for objects that aren't moving anywhere near the speed of light. Instead, the following is true even for objects moving at relativistic speeds:

$$\vec{F} = \frac{d\vec{p}}{dt} \approx \frac{\Delta\vec{p}}{\Delta t}$$

Note that the right side of this is only approximately true, when Δt is very small. In recitation, you'll have used this to adjust and keep track of total momentum:

$$\vec{p} = \vec{p} + \vec{F}\Delta t$$

But how do we find the velocity from the momentum alone (we can't just use $\vec{v} = \vec{p}/m$)? You may remember the general form of momentum in terms of mass and velocity (simplifies to $\vec{p} = m\vec{v}$ at low speeds):

$$\vec{p} = m\vec{v}\gamma$$

Let's write this equation in one dimension (no vectors required) and expand out the factor of γ:

$$p = \frac{mv}{\sqrt{1 - \frac{v^2}{c^2}}}$$

Now we have an equation with a single unknown (v, assuming we already have the momentum and mass of the object). That means we can do some algebra! The following are the steps taken to solve for v here:

$$p^2 = \frac{m^2v^2}{1 - \frac{v^2}{c^2}}$$

$$p^2\left(1 - \frac{v^2}{c^2}\right) = m^2v^2$$

$$p^2 = m^2v^2 + \frac{p^2v^2}{c^2}$$

$$p^2 = v^2(m^2 + \frac{p^2}{c^2})$$

$$v^2 = \frac{p^2}{(m^2 + \frac{p^2}{c^2})}$$

$$v = \frac{p}{\sqrt{m^2 + \frac{p^2}{c^2}}}$$

And if we simplify by dividing both the numerator and denominator by a factor of m:

$$v = \frac{p/m}{\sqrt{1 + (\frac{p}{mc})^2}}$$

This is how you can convert directly from momentum to velocity, while still taking special relativity into account (you can check this using a calculator – no matter how high you set the momentum, the speed of the object will never surpass the speed of light!).

And that concludes our foray into the world of special relativity! Again, we needed this for recitation, but barring what you see in lecture, you will not need to know any of this on your exams for this course.